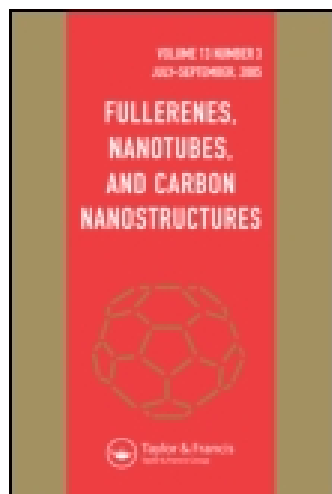


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Synthesis of Boron-Doped C₆₀ Film Using Plasma-Assisted Thermal Evaporation Technique and its Electrochemical Characterizations

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Boron doped C₆₀ (B:C₆₀) thin films were synthesized by a plasma-assisted thermal evaporation system. Scanning electron microscope (SEM), Raman spectroscopy, Fourier transform infra red (FTIR) and X-ray photo electron spectroscopy (XPS) were used to investigate the morphology and structural characteristics of the films, respectively. The electrochemical characterization of B:C₆₀ films was then introduced as a coating layer for silicon film anodes in a lithium ion battery. Typical electrochemical measurements such as cyclic voltammetry (CV) and impedance spectroscopy were used to characterize the B:C₆₀ coated silicon electrodes. CV profiles implied that B:C₆₀ coating layer could act as a passive layer with respect to the insertion and extraction of lithium ions into the silicon film electrode. Moreover, impedance results showed that the coating layer contributed to the smaller charge transfer resistance at the electrode/electrolyte interface.

Keywords Boron, doping, C₆₀, coating, electrochemical

Introduction

In 1985, Fullerene C₆₀ was invented for the first time by Kroto and Smalley using in arc discharge method (1). Subsequently, C₆₀ and its derivatives have been developed in various applications including solar cells (2), light emitting diodes (3) and hydrogen storage (4) due to the unique structure and properties. Intrinsic C₆₀ films are semiconductor materials possessing a band-gap of 1.5 eV and an electrical resistivity of 10³–10⁸ Ω.cm at room temperature. In order to improve the conductivity of C₆₀ films, the insertion of boron atom into the C₆₀ lattices can be introduced. The preparation of boron doped C₆₀ films was reported by Zou et al. (5,6); however, there was no applications proposed in their works.

In this present study, the boron doped C₆₀ films were synthesized by the plasma-assisted thermal evaporation technique of C₆₀ powder. The electrochemical characterization of the doped C₆₀ film was then conducted by using it as a coating layer for silicon film anodes in lithium secondary batteries. Moreover, the effect of insertion of boron into C₆₀ film on the structural and electrochemical characteristics of coated silicon electrodes was investigated.

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Experimental

Synthesis of Thin Films

The B:C₆₀ thin films were grown in a radio frequency (rf)-powered plasma-assisted thermal evaporation system at temperature of 150°C with argon as the carrier gas. As the carbon source, the C₆₀ powder was loaded into a tungsten boat. The substrate was located on the steel holder and was about 10 cm from the boat. The used substrate was the silicon wafer for the structural studies and silicon film on Cu foil for the electrochemical characteristics. Initially, the pressure inside the chamber was evacuated to the base pressure of 10⁻⁵ Torr using a rotary pump. Then the mixture gas of argon and diborane (B₂H₆) entered the chamber with a flow rate of 30 sccm to reach a working pressure of 2.5.10⁻² Torr. The flow was controlled by a mass flow controller (MFC). The radio-frequency generator was activated to generate the argon plasma. At the same time, the C₆₀ powder was evaporated by thermal heat input. Therefore, the C₆₀ vapor in the argon plasma environment could be deposited in the form of thin film onto the substrate, and the boron atoms could be inserted into fullerene film lattices. The rf-plasma power of 20 and 100 W was used during deposition process. For comparison, the pristine C₆₀ film was also deposited by thermal evaporation technique of C₆₀ powder. The evaporation was done under the same conditions as that of B:C₆₀ film without feeding the argon gas.

Structural and Morphological Characterizations

The surface morphology and structure of the thin films was examined by scanning electron microscope (SEM), Raman spectroscopy (Nicolet Almega XR, Thermo Electron Corporation with 514.5 Ar laser), Fourier transform infra red spectroscopy (FTIR, Nicolet Almega XR, Thermo Electron Corporation) and X-ray photo electron spectroscopy (XPS, VG Scientific ESCALAB 200R), respectively. The bulk resistivity of the thin film samples was measured by four point probe analyzer.

Electrochemical Measurements

Electrochemical characteristic was studied by cyclic voltammetry (CV) measurements under scan rate of 0.05 mVs⁻¹ within cut-off voltage of 0–3 V. For CV tests, half cells were then fabricated in a dry room. The half-cells were fabricated by sandwiching a polyethylene (PE) separator between the B:C₆₀ coated Si film electrode (as the working electrode) and Li metal (as the counter electrode). The half-cell (2 cm × 2 cm) was then sealed in an aluminized polyethylene bag. The electrolyte was 1 M LiPF₆ in a mixture of ethylene carbonate (EC), ethyl methyl carbonate (EMC) and dimethyl carbonate (DMC) (1:1:1 by volume). Electrochemical impedance spectroscopy was conducted using Zahner electric instruments with frequency range between 0.01 Hz and 10⁶ Hz and amplitude ratio of 10 mV.

Results and Discussion

Figure 1a presents the SEM image of pristine C₆₀ film synthesized by thermal evaporation technique without using argon plasma. From the surface view, it can be seen that the film consist of many grains with different sizes due to the evaporation of C₆₀ powder. In contrast, the B:C₆₀ thin film prepared by the plasma-assisted thermal evaporation technique

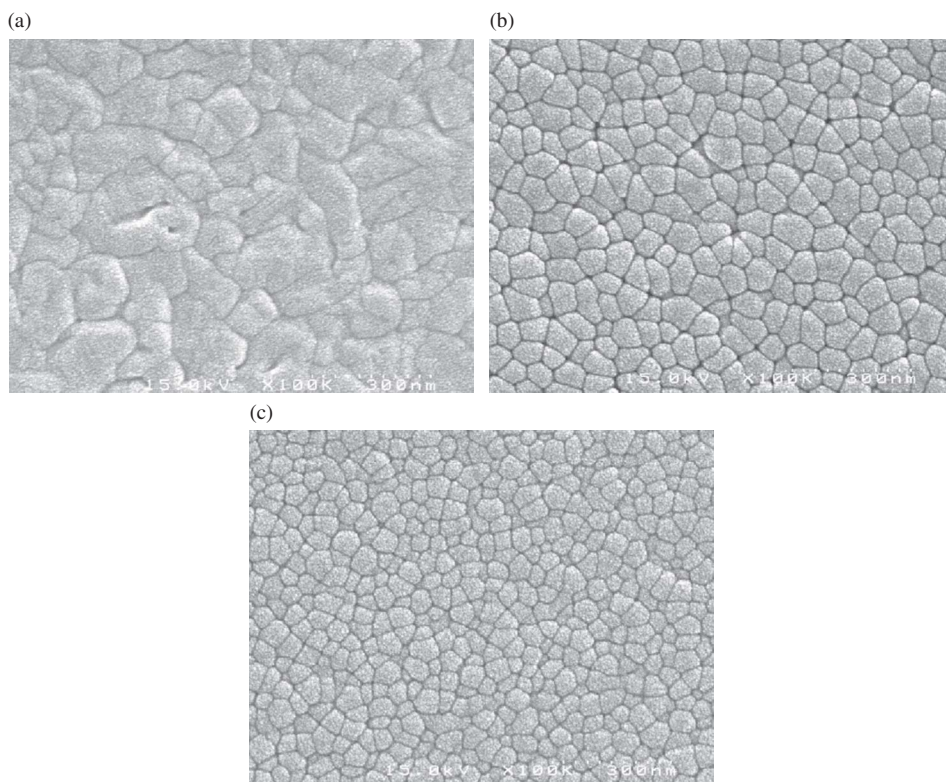


Figure 1. Surface view SEM image of (a) pristine C_{60} and $B:C_{60}$ thin film synthesized at plasma power of (b) 20 and (c) 100 W.

shows a smaller grain size with more uniform shape, as depicted in Figures 1b and 1c. It seems that the insertion of boron atoms assisted with the argon plasma treatment during the film deposition contributes to the smaller grain size of $B:C_{60}$ film. It can be seen also that the smaller grain size is obtained by increasing the plasma power from 20 to 100 W. The role of argon plasma environments is quite critical to create more uniform structure with smaller cluster size in comparison with that of pristine films.

The surface composition of the thin films was determined by XPS analysis, as shown in Table 1. As seen, the boron concentration in the thin films was increased by increasing the plasma power from 20 to 100 W, indicated that the amount of boron doped into the C_{60} films can be controlled by plasma power during film deposition. The films contain a small amount of oxygen on their surface as contaminants. However, the concentration of oxygen is still tolerable at lower than 5%. As shown in Table 1, the surface resistivity of the thin films was measured by four-point probe instruments. It is apparent that the resistivity of C_{60} sample can be improved due to introduction of boron atoms into the C_{60} lattices. In other words, the conductivity of the C_{60} films was enhanced by the presence of boron doping.

Figures 2a shows the C 1s XPS spectra of pristine C_{60} and $B:C_{60}$ films. The C 1s binding energies of the pristine C_{60} and $B:C_{60}$ films are determined as 284.2 and 285.2 eV. The binding energy is shifted to the higher value due to the presence of stronger bonding between carbon molecules. The full width at half maximum (FWHM) of C 1s for the $B:C_{60}$

Table 1
Composition of thin film samples as determined by XPS

Sample	Resistivity ($\Omega\cdot\text{cm}$)	Atomic concentration		
		B	C	O
Pristine C_{60}	6.67×10^5	0	98.75	1.25
B: C_{60} (20 W)	2.25×10^3	1.24	97.4	1.36
B: C_{60} (100 W)	1.17×10^3	3.52	94.48	1.5

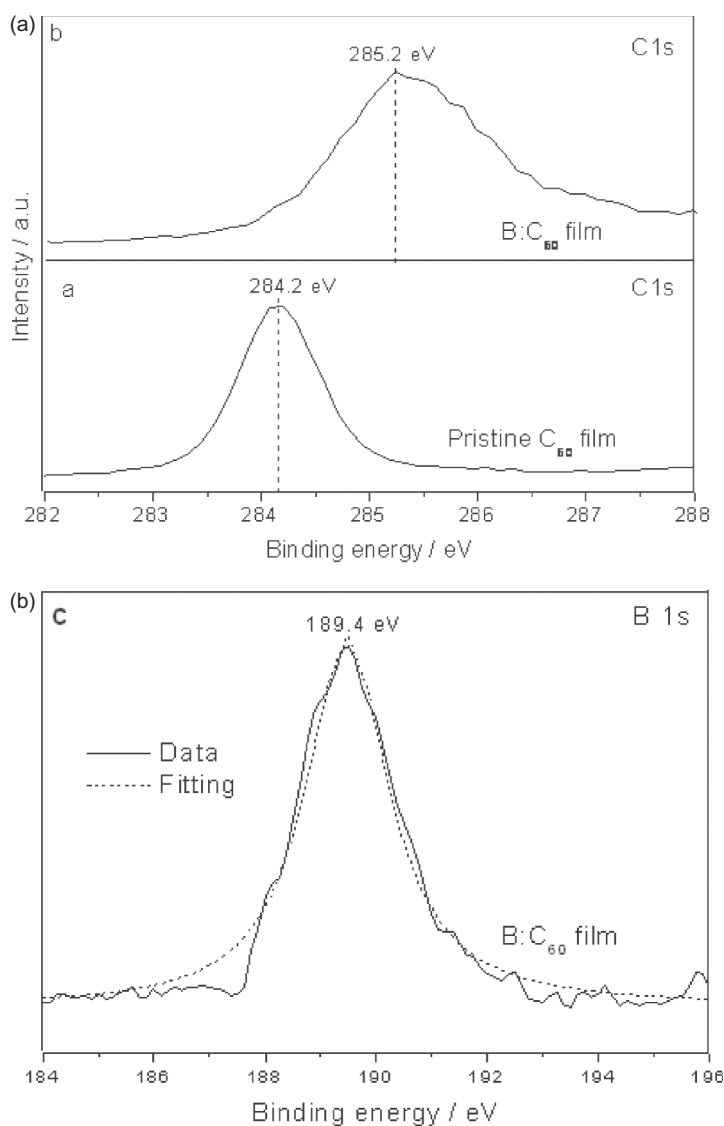


Figure 2. XPS spectra of (a) C 1s and (b) B 1s for pristine and B: C_{60} films.

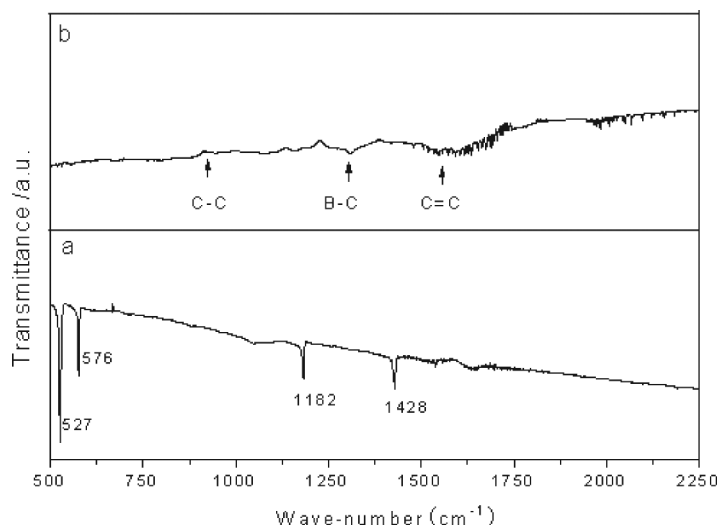


Figure 3. FTIR spectra of as deposited (a) pristine C_{60} and $B:C_{60}$ thin film.

films is determined as 1.58 eV. This value is obviously larger than that of pristine form (0.96 eV). This higher FWHM value and broader spectrum of the $B:C_{60}$ films indicate that the linking among the fullerene molecules has taken place during the deposition process. The presence of boron atoms in the fullerene lattices is evidenced by the B 1s XPS spectra of $B:C_{60}$ film, as shown in Figure 2b. The peak centered at 189.4 eV is an indication of the bonding between boron and carbon in the fullerene film, which are consistent with those found in the literature (7,8).

Figure 3 shows the FTIR spectra of pristine C_{60} film deposited by evaporation technique without argon plasma treatment and $B:C_{60}$ film synthesized within argon plasma environment. For pristine C_{60} film, there are four dominant peaks detected at wave-numbers 511, 527, 1182 and 1428 cm^{-1} . However, those four peaks are not found in the case of $B:C_{60}$ film, in which there are many new peaks related to the vibration of bonding states of C-C, C=C and B-C. Therefore, the bonding between boron and carbon in the $B:C_{60}$ film can be detected due to the presence of weak vibration of B-C bond located at wave-number of 1260 cm^{-1} .

Figure 4 shows the Raman spectra of the pristine C_{60} film deposited by the evaporation technique without argon plasma and the $B:C_{60}$ film synthesized with plasma power of 20 and 100 W. It can be seen that a strong peak at Raman shift of 1468 cm^{-1} , known as the pentagonal pinch mode, is observed in the case of pristine C_{60} film (7,8). This pentagonal pinch mode is gradually separated and broadened, but its existence is still recognized in the case of $B:C_{60}$ films. This broadened peak in the vicinity of pentagonal pinch is the main indication of the typical fullerene C_{60} structure, which cannot be found in other structure of carbon such as graphite, diamond or carbon nanotubes (9–12).

The electrochemical characteristic of pristine C_{60} and $B:C_{60}$ film as a coating layer for silicon film is initially conducted by the cyclic voltammetry (CV) measurements. The aim of CV tests is to investigate the reversible reaction between lithium ions and silicon film as active materials. The CV curves of pristine coated silicon film for the first cycle under a scan rate of 0.05 mVs^{-1} are shown in Figure 5a. In the case of pristine C_{60} coated silicon film, there are no anodic peaks (in the positive current direction) detected in the

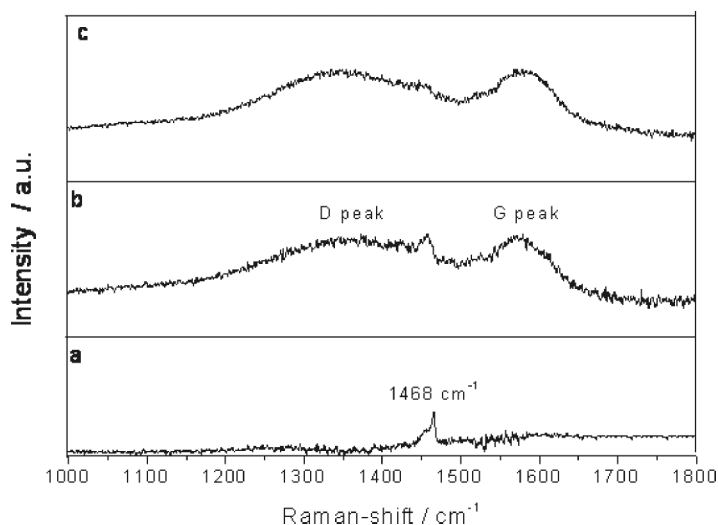


Figure 4. Raman spectra of as deposited (a) pristine C_{60} and B: C_{60} thin film at plasma power of (b) 20 and (c) 100 W.

CV profiles. Since the anodic peaks are representing the insertion of lithium ions into silicon anodes, it indicates that the pristine C_{60} film is blocking the transport of lithium ions into silicon anodes during the charge-discharge tests. The three cathodic peaks (in the negative current direction) located at 1.49, 1.85 and 2.05 V might refer to the reaction of the silicon anodes with the electrolyte, since we know from our previous work that the reaction between lithium ions and silicon is indicated by the cathodic peaks at voltage range between 0 and 0.4 V (13).

Figure 5b shows the CV curves of B: C_{60} coated silicon film synthesized at plasma power of 100 W for the first cycle under scan rate of 0.05 mVs^{-1} . As seen, there is two pair of cathodic and anodic peaks found at the negative and positive current direction, respectively. The cathodic peaks which represent the extraction of lithium ions from silicon anodes are observed at 0.02 and 0.22 V, and the anodic peaks which refer to the insertion of lithium ions into silicon anodes are found at 0.38 and 0.5 V (13). It seems that the B: C_{60} coating layer is quite effective to preserve the insertion and extraction of lithium ions to the silicon anodes during charge-discharge tests and also preventing the side reaction with the electrolyte. In addition, the current intensity observed in the CV curve of B: C_{60} coated silicon anodes is much higher than that of pristine C_{60} , indicating that the B: C_{60} film is more conductive for the transport of lithium ions since the current intensity is proportional to the amount of lithium ions passed through the silicon/electrolyte interface. The CV profiles of B: C_{60} coated Si film synthesized at plasma power of 20 W is similar to that of sample synthesized at 100 W. Based on the CV results, the effect of boron doping assisted by the generation of argon plasma during the coating process is significant to preserve the reversible reaction between lithium ions and silicon anodes, since it can form a more conductive layer with respect to lithium ions than that of pristine film.

From the Nyquist plots of fullerene C_{60} coated silicon electrode, which is depicted in Figure 6, we can observe that the shortest diameter of semi circle is shown by the B: C_{60} coated silicon film synthesized at plasma power of 100 W. Since the diameter of the semicircle is equivalent with the charge transfer resistance, the coating layer provides

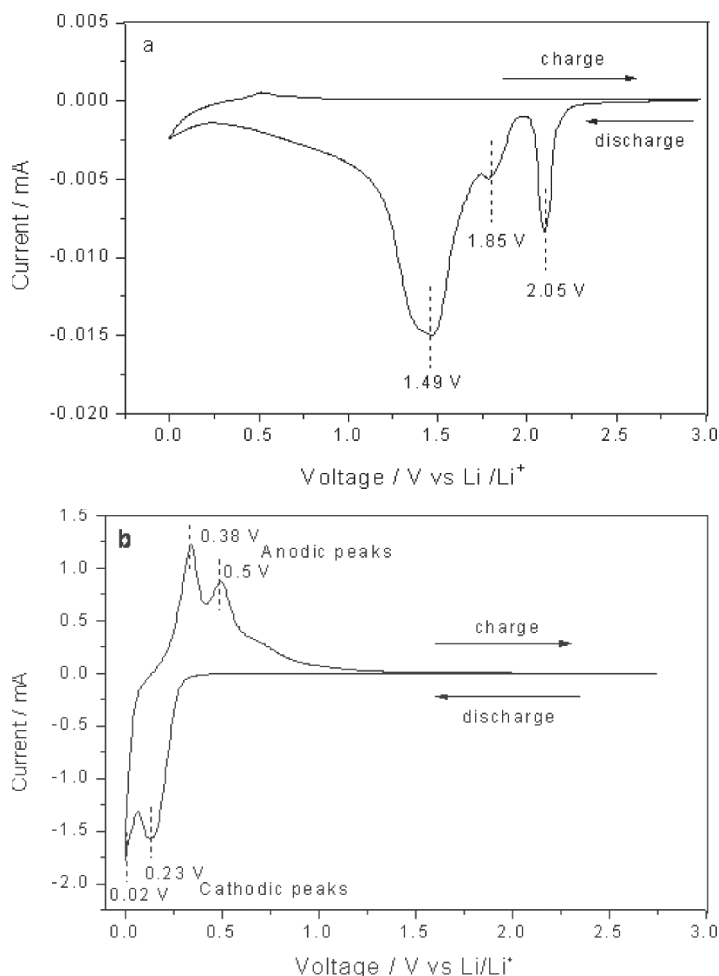


Figure 5. Cyclic voltammetry curves of (a) pristine coated Si and (b) B:C₆₀ coated Si electrodes synthesized at plasma power of 100 W for the first cycle under scan rate of 0.05 mVs⁻¹ between 0–3 V.

a better interfacial contact with the electrolyte. This is due to the improvement of surface conductivity after the B:C₆₀ coating.

Conclusions

Boron doped fullerene (B:C₆₀) thin film has been successfully synthesized by plasma assisted thermal evaporation technique. The presence of boron atoms in the fullerene lattices was confirmed by SEM image, Raman, XPS and FTIR analysis. B:C₆₀ film was then utilized as a coating layer for silicon anodes in lithium secondary batteries. CV results indicated that B:C₆₀ film was quite effective to preserve the reversible reaction between lithium ions and silicon. In comparison with the pristine C₆₀ films, the B:C₆₀ films can provide a conducting path for the transport of lithium ions at the electrode/electrolyte interface. Hence, it contributes to the smaller charge transfer resistance shown by the B:C₆₀ coated silicon films.

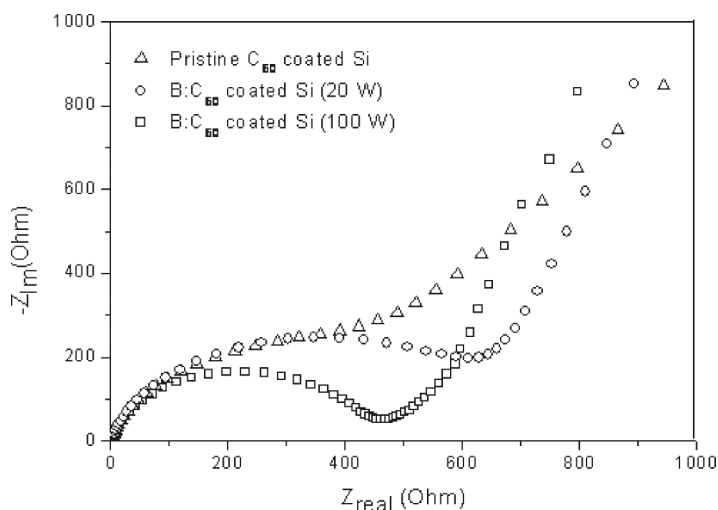


Figure 6. Nyquist impedance spectra of (a) pristine C_{60} and B: C_{60} thin film at plasma power of (b) 20 and (c) 100 W under frequency range between 0.01 Hz and 10^6 Hz.

Acknowledgment

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